

UMHackathon 2025: Preliminary Round Proposal

Domain 2: Quantitative Trading

Team: Sleep Not Found (235)

1.0 Executive Summary

We propose an **alpha trading system** for Cryptocurrency markets that combines **Hidden Markov Models (HMM)**, **XGBoost**, and **LSTM networks** to identify market regimes and generate high-conviction trading signals. Our backtesting shows an annualized return of 288% with a Sharpe ratio of 4.63, significantly outperforming benchmark requirements while maintaining controlled risk exposure.

2.0 Problem Statement

The cryptocurrency market is characterized by **extreme volatility** and **non-stationary behavior**. Price movements are heavily influenced by diverse and unpredictable factors such as global news events, market sentiment, regulatory developments, and large-scale whale activity. These influences cause rapid shifts in market dynamics, making it difficult for traditional analytical methods to maintain predictive accuracy.

Most existing trading models are designed under the assumption of stationarity and rely heavily on technical indicators like RSI, MACD, and moving averages. These indicators operate on fixed time windows and static rules, which fail to adapt in real-time. As a result, they often produce misleading or delayed signals in fast-changing environments.

Furthermore, conventional models tend to ignore **market regime shifts**—the transitions between bull, bear, accumulation, and distribution phases. This oversight leads to poor contextual understanding and inaccurate decision-making, as all data is treated uniformly regardless of underlying conditions.

3.0 Our Value Proposition

We propose a **hybrid, regime-aware trading strategy** that **integrates statistical models with machine learning** to adapt dynamically to current market conditions. Our system is designed to **detect regime shifts, generate high-confidence trading signals, and manage risk proactively**.

1. Model Architecture

- **Hidden Markov Models (HMM)**

Enables detection of latent market regimes (bullish, bearish, neutral) through probabilistic state transitions. By uncovering hidden trends, the model helps avoid costly trades during uncertain phases.

- **XGBoost (Extreme Gradient Boosting)**

Performs feature ranking and selection. It highlights which indicators—such as whale activity, inflow/outflow metrics, or volatility—are most predictive under current conditions.

- **LSTM (Long Short-Term Memory Networks)**

Specialized in learning sequential dependencies in time-series data. LSTM is used to forecast short-term price trends by capturing temporal patterns in historical price behavior.

- **Kalman Filtering**

Reduces noise while preserving essential price action patterns

2. Data Sources

To provide a comprehensive view of market conditions, we construct a multi-dimensional dataset composed of:

- **Inflow/Outflow Data:** Captures market supply-demand dynamics
- **Close Price Data:** Provides essential technical foundation for pattern recognition
- **Exchange Whale Ratio:** Detects large-scale market movements before price action
- **Combined Dataset:** Creates multi-dimensional view that individual indicators miss

3. Regime-Based Execution

The system adjusts its trading behavior based on the identified regime:

- **Bullish Regime:** Enter LONG positions when ML model predicts upward movement
- **Bearish Regime:** Enter SHORT positions when ML model predicts downward movement
- **Neutral Regime:** Remain in cash or reduce exposure to minimize risk
- **Risk Controls:** Position sizing adjusts based on volatility with strict drawdown limits

4. Risk Management Framework

We employ robust risk mitigation techniques:

- **Trading Cost Optimization:** Algorithm accounts for 0.06% exchange fees
- **Drawdown Control:** Position sizing adjusts based on volatility to limit maximum drawdown
- **Signal Filtering:** Only highest-conviction signals executed to maintain quality over quantity

4.0 Competitive Advantages

- **Superior Statistical Foundation:** Multi-model approach compensates for individual weaknesses
- **Adaptive Market Intelligence:** System automatically adjusts to changing market conditions

- **Practical Implementation:** Fast execution with emotional factors removed
- **Risk Management:** Stringent controls with maximum drawdown limited to 33%

5.0 Performance Metrics

- **Annualized Return:** 288%
- **Sharpe Ratio:** 4.63 (≥ 1.8)
- **Maximum Drawdown:** 33.15% ($\geq -40\%$)
- **Trading Frequency:** 9.12% ($\geq 3\%$)
- **Total Trades:** 33 (6 buys, 11 sells)

6.0 Conclusion

Our hybrid trading system combines HMM, XGBoost, and LSTM models to deliver exceptional results in cryptocurrency markets. With a 288% annualized return and 4.63 Sharpe ratio, we've created a strategy that:

- Adapts to changing market conditions
- Eliminates emotional trading decisions
- Provides consistent returns in both bull and bear markets
- Maintains controlled risk with a 33% maximum drawdown

This systematic approach offers a scalable solution for generating alpha in cryptocurrency markets while meeting all project requirements.